Abhinav Sharma

Cybersecurity Researcher · Agentic AI for Security Operations

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PROFILE

Cybersecurity researcher and educator with a PhD focused on enhancing SIEM capabilities through threat analytics and intelligence. My research centres on agentic AI and multi-agent cognitive architectures for autonomous, proactive security operations — bridging applied SOC practice with frontier-AI methods. Author of six peer-reviewed papers on AI-driven security operations, including work on fine-tuning open-source LLMs for SOC analysis, explainable alert triage, and insider-threat detection.

SELECTED PUBLICATIONS & RESEARCH

1. **ALUSKORT: A Hierarchical Multi-Agent Cognitive Architecture for Autonomous Security Operations.** International Journal of Applied Mathematics (IJAM), Vol. 38, No. 5s, 2025. SCOPUS · Q3
2. **Hallucinations of the Generalist, Brittleness of the Expert: A Study on Fine-Tuned Open-Source LLMs for SOC Analysis.** International Journal of Science and Engineering (IJSE), Vol. 11, Issue 02, Sept. 2025. DOI: 10.53555/1vhsgg43.
3. **Deconstructing AI Confidence: An Explainable Framework for Articulating Alert-Triage Confidence in AI-Enabled SOCs.** Journal of The Institution of Engineers (India): Series B, Springer. COMMUNICATED
4. **An Epistemic Recalibration of the Security Operations Center: Agentic AI as the Vanguard of Proactive Cyber Defense.** Intl. Conf. on Artificial Intelligence, Computer and Electrical Engineering (IACE-2025), Guru Kashi University, in collaboration with ISTE Delhi & McGill University, Canada, Sept. 2025.
5. **Insider Threats as Digital Corporate Fraud: Lessons from the KiranaPro Breach and the Promise of AI-Driven SOCs.** FINCON 2025, International Financial Security & Management Conference, National Forensic Sciences University, Gandhinagar, Aug. 2025.
6. **Empowering SOCs: A Conceptual Analysis of Agentic AI's Transformative Potential in Cybersecurity Operations.** 5th Intl. Conf. on Computational Methods in Science & Technology (ICCMST-2025), CGC Landran, Mohali, Aug. 2025.

DOCTORAL RESEARCH

PhD, Cyber Security — Himachal Pradesh University, Shimla

Enhancing SIEM Capabilities with Threat Analytics and Intelligence: The PICERL Approach

Supervisors: Dr. Jawahar Thakur (Professor, Dept. of Computer Science, HPU Shimla); Dr. T. P. Sharma (Associate Professor, Dept. of Computer Science & Engineering, NIT Hamirpur).

PROFESSIONAL EXPERIENCE

Assistant Professor, Computer Applications — 2018 – Present

Government College Bhoranj (Tarkwari), Hamirpur, Himachal Pradesh

- Teach computer applications and cybersecurity foundations; lead independent research on agentic AI for security operations alongside teaching.

Programmer — Feb 2018 – Oct 2018

Himachal Pradesh Administrative Tribunal, Shimla

Assistant Programmer — Sep 2016 – Feb 2018

Hon'ble High Court of Himachal Pradesh, Shimla

- Supported statewide rollout of the Case Information System (CIS) under the e-Courts programme at District & Sessions Court, Shimla.

State Project Coordinator, MDMS-MIS & ARMS — Apr 2016 – Sep 2016

Directorate of Elementary Education, Government of Himachal Pradesh, Shimla

- Coordinated statewide implementation of the MDMS–ARMS reporting system and departmental web operations.

EDUCATION

Advanced Program in Cyber Security & Cyber Defense
IIT Kanpur

M.Tech, Computer Science
Dept. of Computer Science, Himachal Pradesh University, Shimla

B.Tech, Information Technology
University Institute of Information Technology (UIIT), HPU, Shimla

CAPSTONE & PROJECTS

Implementing a Security Operations Centre (SOC)
Capstone, Advanced Program in Cyber Security & Cyber Defense, IIT Kanpur.

ALUSKORT — Multi-Agent SOC Architecture
Hierarchical multi-agent cognitive architecture for autonomous security operations (published research prototype).

TECHNICAL FOCUS

SIEM & Threat Analytics · SOC Automation · Threat Intelligence · Incident Response (PICERL) · Agentic AI · Multi-Agent Systems · LLMs for Security · LLM Fine-Tuning · Explainable AI · Alert Triage · Insider-Threat Detection · Adversarial Reasoning

AFFILIATIONS & ACTIVITIES

- Associate Member, The Institution of Engineers (India) — Computer Division.
- Editor & Designer, Himachal Pradesh State Legal Services Authority biannual newsletter (2016–2018).

“Perform all work carefully, guided by compassion.” — Ved Vyasa, Bhagavad Gita